M2 Literature and Credibility Plan

Differentiable Self Calibration for Fourier Ptychography

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1 Prior Work

Work and question	Model form and assumptions	Verification or validation evidence	What we adopt or avoid
[1] Eckert et al. (2018) How can illumination angles be self-calibrated from FP measurements?	Fourier-circle detection initializes angles; spectral correlation refines them within iterative FP. Assumes coherent thin-object FP and basic setup knowledge.	Corrects small and large misalignment in simulated and experimental 2D/3D FP with LED, galvo, and quasi-dome sources.	Adopt angle calibration and a fixed-geometry baseline. Omit image-processing initialization. Our rigid dx, dy, theta model is intentionally narrower.
[2] Guzzi et al. (2023) Can automatic differentiation support rapid prototyping across imaging inverse problems?	Differentiable operators, gradient optimization, and GPU execution for X-ray inverse problems; forward operations and loss must be differentiable.	Case studies include super-resolution, ptychographic formulations, and tomography. Evidence supports method design, not our LED-pose model.	Adopt modular autodiff and gradient optimization. Verify derivatives independently with centered finite differences.
[3] Chen et al. (2025) Can one differentiable FP model jointly recover the object, misalignment, aberrations, and low-quality data effects?	Differentiable Fourier optics with optimizable illumination pose, regularized Zernike pupil terms, and data-quality variables.	Uses experiments and controlled simulations with pose/aberration truth; reports parameter recovery and reconstruction gains.	Adopt joint physical parameters, synthetic truth, and aligned error. Keep the baseline pose-only; reserve aberrations for the extension.

Position relative to prior work. For one 256 x 256 synthetic object and a centered 7 x 7 LED subset, does estimating rigid array translation and rotation reduce aligned object NRMSE versus fixed geometry? [1] motivates angle calibration, [2] autodiff, and [3] the broader joint model. We isolate pose before adding aberrations.

2 Validation Data and Trusted References

Asset	Access and terms	Fields and units	Selected slice and use
Synthetic reference generator	Team-controlled; seed 20261004. Version generator, parameters, seed, and hashes. No external restrictions.	Complex object amplitude (dimensionless), phase (rad); dx and dy (mm); theta (degree); wavelength (nm); intensity (normalized); pupil (dimensionless complex).	256 x 256 object; 530 nm; NA 0.10; centered 7 x 7 LEDs (49 frames); 10 matched pose/noise seeds. Exact truth gives aligned NRMSE and pose errors.
UConn Smart Imaging Lab Fourier-Ptychography repository [4]	Retrieved Oct 4, 2026. Code: GPL 2.0 plus attribution notice. Dataset terms are unstated; retain attribution and limit use to course research unless clarified.	imlow HDR intensity (detector units not specified); theta (degree); wlength (m); xint and yint (mm); z defocus (m); optional aberration (dimensionless complex pupil/phase data).	HE_blue.mat: 141 x 141 frames. Use centered 7 x 7 of the 15 x 15 stack (49 frames): Nikon 2X, NA 0.1, 1.845 micrometer sample pixel, 4 mm pitch, 90.88 mm LED distance.

Experimental Limits and Claims

Opened asset statement. Raymond Yao opened HE_blue.mat in MATLAB on October 4, 2026 and confirmed low-resolution frames plus rotation, wavelength, offsets, defocus, and aberration metadata; data_description.txt lists the same fields and geometry.

What would directly validate the model. Direct NRMSE validation needs independent complex-object truth and surveyed LED coordinates; neither is supplied. Thus synthetic tests validate recovery, while real data support only convergence, consistency, and plausible pose. Residual reduction alone does not prove accuracy.

Repository documentation. data/README.md will record URL, date, terms, opener, variables/units, selected indices/slice, preprocessing, seed, and checksum.

Stable asset links: UConn repository; data description; license.

3 Verification Plan

Purpose. Run these checks before validation. The centered finite-difference comparison remains the M1 verification anchor.

Check	What it tests	Specific failing result	Record
Centered finite-difference gradient check	Autodiff derivatives of loss with respect to dx, dy, theta on a 64 x 64 case. Test h = 10^-2, 10^-3, 10^-4 after order-one scaling.	Fail if minimum relative error $\|g_{AD} - g_{FD}\| / \max(1, \|g_{AD}\|, \|g_{FD}\|) > 10^{-4}$ for any variable, or no decrease-to-roundoff trend appears.	results/verification/gradient_check.csv and gradient_check.png
Known-answer zero-misalignment limit	Forward-model bookkeeping and optimizer behavior for zero pose/noise with matched generating and reconstruction pupils.	Fail if true-state normalized residual > 10^-10, recovered pose > 0.01 mm or 0.01	results/verification/zero_pose.json and zero_pose_summary.md

		degree, or joint versus fixed aligned NRMSE differs by > 10^-3.	
Spatial refinement and run sufficiency	QoI stability across 64, 128, 256 grids and 10, 20, then 40 paired pose/noise trials.	Fail if median NRMSE gain shifts > 2 percentage points from 128 to 256, or its 95% bootstrap CI half-width stays > 2 points after 40 trials.	results/verification/refinement.csv and run_sufficiency.csv

Decision Rules

Order of execution. Run gradient, known-answer, then refinement checks. Any failure blocks the headline comparison until a clean rerun passes.

Headline comparison. With matched seeds, report both NRMSEs, paired difference, pose errors, and runtime. Success requires at least 5% lower median aligned NRMSE; lower residual without lower truth-based NRMSE fails.

Reproducibility. Record commit, configuration, seed, grid, stopping rule, software version, and runtime. Regenerate plots from saved CSV/JSON files.

4 Ranked Uncertainty Register

Rank	Source and kind	Range and basis	Expected QoI effect	First diagnostic
1	Unmodeled pupil aberration - model form	Ideal pupil versus supplied aberration field; extension uses low-order Zernike terms [3,4].	May mimic pose, lowering residual while biasing pose and object NRMSE.	Compare ideal, supplied, and jointly estimated low-order pupils on identical frames.
2	True LED-array pose - input/parameter	dx, dy in [-2, 2] mm; rotation in [-2.5, 2.5] degree, following [3].	Larger errors raise baseline NRMSE and may expose nonconvex failures.	Paired sweep over translation and rotation while holding object and noise fixed.
3	Measurement noise and dynamic range - input/data	0, 0.5, 1, and 2% peak-normalized noise; compare Gaussian and Poisson-like synthetic noise.	Raises NRMSE/pose variance; dark-field frames may distort intensity-squared loss.	Noise sweep with identical pose; compare intensity and amplitude residual choices.
4	Rigid-array assumption - model form	Global pose versus per-LED perturbations of 0, 0.10, 0.25 mm RMS below the 4 mm pitch [4].	Rigid fit may recover the mean yet leave structured residuals/artifacts.	Generate per-LED perturbations and map residual versus LED index and illumination angle.
5	Grid and interpolation - numerical	64, 128, and 256 object grids; identical physical field and illumination geometry.	Coarse grids can bias spectral shifts, NRMSE gain, and pose.	Refinement run; require less than 2 percentage-point QoI change from 128 to 256.
6	Initialization and stopping - numerical	10 matched starts; 150/300/600 iterations; learning rates 0.5x/1x/2x baseline.	Nonconvex basins can change pose and NRMSE; one run is insufficient.	Multi-start sweep and convergence traces; increase runs until the planned interval criterion is met.

Why rank 1 comes first. Pupil aberration can mimic geometry in the same residual, so a pose-only fit may optimize well but recover the wrong cause, invalidating both NRMSE and interpretation.

5 Graduate Extension

Part	Plan
Need	Estimate low-order pupil aberrations with LED pose to test whether NRMSE gains are geometric or aberration compensation.
Literature	Chen et al. [3] jointly optimize illumination misalignment and Zernike pupil terms, tested with synthetic truth and experiments.
Dependency	Requires the pose-only model, verified gradients, aligned NRMSE, and synthetic generator; adds pupil parameters and regularization.
M3 prototype	Add defocus and one astigmatism term to the 64 x 64 case. Pass if FD versus autodiff error < 10^-4, nonzero coefficients are within 10%, and pose degrades by < 0.02 mm/0.02 degree.

References

[1] Eckert, R., Phillips, Z. F., and Waller, L. (2018). Efficient illumination angle self-calibration in Fourier ptychography. Applied Optics, 57(19), 5434-5442. https://doi.org/10.1364/AO.57.005434

[2] Guzzi, F., Gianoncelli, A., Bille, F., Carrato, S., and Kourousias, G. (2023). Automatic Differentiation for Inverse Problems in X-ray Imaging and Microscopy. Life, 13(3), 629. https://doi.org/10.3390/life13030629

[3] Chen, N., Wu, Y., Tan, C., Cao, L., Wang, J., and Lam, E. Y. (2025). Uncertainty-aware Fourier ptychography. Light: Science and Applications, 14, 236. https://doi.org/10.1038/s41377-025-01915-w

[4] Smart Imaging Lab, University of Connecticut. Fourier-Ptychography Datasets and Codes. Retrieved October 4, 2026. https://github.com/SmartImagingLabUConn/Fourier-Ptychography